

BEN AGRO

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Bringing research to real world robots

Work Experience

Tesla AI - Machine Learning Engineer → Sr. Machine Learning Engineer

📅 March 2025 - Present

- End-to-end driving policies, from pretraining to RL, under Phil Duan
- Currently focused on pretraining scaling; model, data, optimizer, evaluations, etc.
- Co-lead the development of FSD v14, resulting in a step-change improvement in safety and the first full unsupervised Tesla robotaxi

Waabi - Intern → Researcher I → Researcher II

📅 May 2022 - Present

- Developing novel perception and forecasting systems that enable self-driving trucks across perception, forecasting, behavior simulation, and sensor simulation.
- Rapid development from research ideas to improvements on real autonomous trucks.
- Published work through various papers; see MAD, DIO, UnO, ImplicitO, DeTra, QUaD in the "papers" section below.

Robotics Vision and Learning Lab - Undergraduate Researcher

📅 May 2021 - Sept 2021

- Enabled robots to operate in complex scenarios with learning-based task and motion-planning systems
- Built a simulation environment for development, extended **PDDLStream** with learned stream-scoring and a queue-based algorithm, and deployed the system to a Franka Panda robot
- Published our work in the paper "Learning to Search in Task and Motion Planning with Streams"

Autonomous Space Robotics Lab - Undergraduate Researcher

📅 May 2020 - Aug 2020

- Developed a self-supervised semantic LiDAR segmentation pipeline for autonomous navigation
- Published the paper "Self-Supervised Learning of LiDAR Segmentation for Autonomous Indoor Navigation"

Publications and Preprints

MAD: MEMORY-AUGMENTED DETECTION OF 3D OBJECTS

Ben Agro, Sergio Casas, Patrick Wang, Thomas Gilles, Raquel Urtasun
CVPR 2025

Top online non-ensemble method on the Waymo Open Detection Challenge

DIO: DECOMPOSABLE IMPLICIT 4D OCCUPANCY-FLOW WORLD MODEL

Christopher Diehl, Quinlan Sykora, **Ben Agro**, Thomas Gilles, Sergio Casas, Raquel Urtasun
CVPR 2025

DeTra: A UNIFIED MODEL FOR OBJECT DETECTION AND TRAJECTORY FORECASTING

Sergio Casas*, **Ben Agro***, Jiageng Mao*, Thomas Gilles, Alexander Cui, Thomas Li, Raquel Urtasun
ECCV 2024

UNO: UNSUPERVISED OCCUPANCY FIELDS FOR PERCEPTION AND FORECASTING

Ben Agro, Quin Sykora, Sergio Casas, Thomas Gilles, Raquel Urtasun
CVPR 2024, **Oral (top 0.7% of submitted papers)**

QUAD: QUERY-BASED INTERPRETABLE NEURAL MOTION PLANNING FOR AUTONOMOUS DRIVING

Sourav Biswas, Sergio Casas, Quin Sykora, **Ben Agro**, Abbas Sadat, Raquel Urtasun
ICRA 2024

IMPLICIT OCCUPANCY FLOW FIELDS FOR PERCEPTION AND PREDICTION IN SELF-DRIVING

Ben Agro, Quin Sykora, Sergio Casas, Raquel Urtasun
CVPR 2023, **Highlight**

TOWARD GLOBALLY OPTIMAL STATE ESTIMATION USING AUTOMATICALLY TIGHTENED SEMIDEFINITE RELAXATIONS

Frederike Dümbgen, Connor Holmes, **Ben Agro**, Tim Barfoot
Arxiv Preprint 2023

LEARNING TO SEARCH IN TASK AND MOTION PLANNING WITH STREAMS

Mohamed Khodeir*, **Ben Agro***, Florian Shkurti

IEEE Robotics and Automation Letters 2023

SELF-SUPERVISED LEARNING OF LIDAR SEGMENTATION FOR AUTONOMOUS INDOOR NAVIGATION

Hugues Thomas, **Ben Agro**, Mona Gridseth, Jian Zhang, Tim Barfoot

ICRA 2021

Education

Masters in Computer Science, University of Toronto

Supervised by **Raquel Urtasun**

📅 Feb 2025 - Oct 2025

PhD in Computer Science (Dropout), University of Toronto

Supervised by **Raquel Urtasun**

📅 Sept 2023 - Feb 2025

BASc in Engineering Science, University Of Toronto

Majored in robotics, Cumulative Average = 97%, CGPA = 4.0

📅 Sept 2019 - May 2023

Awards

John Black Aird Scholarship, W.S. Wilson Medal, Ontario Professional Engineers Foundation Gold Medal, Governor's General Silver Medal

📅 May 2024

Awarded to the student with the highest academic standing of any UofT undergraduate

Centennial Senior Project Award

Awarded for the best **undergraduate thesis project**

📅 May 2024

University of Toronto Excellence Award

To fund the research of exceptional UofT undergraduate students

📅 May 2020

University of Toronto Scholar Award

Recognition of UofT's outstanding students at admission

📅 Sept 2019

Governor General's Bronze Medal

Awarded for the highest academic standing in highschool

📅 May 2019

AP National Scholar

Graduated highschool with six AP (university equivalent) credits

📅 May 2019

Personal Projects

BARFT: BUNDLE ADJUSTING NEURAL RADIANCE FIELDS WITH TEMPORAL REGULARIZATION

A framework for training NeRFs with unknown (learned) camera poses.



EXPLAINER OF "TRANSFORMERS AS STATISTICIANS"

Distilling the key ideas behind "**Transformers as Statisticians: Provable In-Context Learning with In-Context Algorithm Selection**"



ZERO-SHOT VIDEO RETRIEVAL WITH VISION LANGUAGE MODELS

A zero-shot video retrieval system that leverages open-source vision-language models



TOWARDS GLOBALLY OPTIMAL STEREO LOCALIZATION (UNDERGRAD THESIS)

An investigation into how to make the problem stereo localization globally optimal, supervised by Tim Barfoot



"CAPTOR" THE AUTONOMOUS DRONE

A custom-built autonomous drone with reliable onboard SLAM and vision-only obstacle avoidance



A version of the popular game Geomety Dash programmed for the original Gameboy hardware

News

Top U of T undergraduate Ben Agro is taking his passion for research into a direct-entry PhD

📅 June 12, 2023